

AMNA NEHAL

MERN Stack Developer | Full-Stack Developer | React.js | Node.js

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PROFILE

Results-driven MERN Stack Developer with hands-on experience building and deploying full-stack web applications using MongoDB, Express.js, React.js, and Node.js. Skilled in RESTful API development, JWT authentication, role-based authorization, and responsive UI design. Experienced in both independent project delivery and collaborative Agile team environments. Currently expanding expertise in AI-powered application development using LangChain, FAISS, and FastAPI. Strong foundation in software engineering principles, MVC architecture, and version control with Git and GitHub.

TECHNICAL SKILLS

Frontend: React.js, TypeScript, JavaScript (ES6+), jQuery, HTML5, CSS3, Bootstrap, Responsive Design, Single Page Applications (SPA), Web Accessibility (WCAG)

State Management: React Context API, React Hooks

Backend: Node.js, Express.js, RESTful API Development, GraphQL, ASP.NET Core MVC, FastAPI

Databases: MongoDB, Mongoose, MySQL, SQL Server, CRUD Operations, Database Schema Design

Authentication: JWT (JSON Web Tokens), Role-Based Authorization, User Authentication & Session Management

Build Tools & Platforms: Git, GitHub, Vite, Postman, VS Code, Vercel, Railway, Render, npm

Methodologies: Agile Development, MVC Architecture, Object-Oriented Programming (OOP), Version Control

AI & Emerging Tech: LangChain, FAISS, Retrieval-Augmented Generation (RAG), LLM Workflows, Document Embeddings, Semantic Search

Documentation: System Architecture Diagrams, Process Flow Diagrams, Technical Reports, API Documentation

EXPERIENCE

Web Developer Intern — CodeAlpha

2023

- Contributed to delivery of responsive web applications, collaborating with team members across frontend and backend tasks
- Coordinated API integration between UI components and backend services, ensuring consistent data flow and system behavior.
- Participated in Agile workflows using Git and GitHub — managing branches, pull requests, and code reviews
- Gained exposure to project tracking and team communication in a professional development environment

Junior Web Developer — Clevertech (Remote, Seasonal)

Jun 2025 – Aug 2025

- Developed and maintained full-stack web applications using React.js, Node.js, and Express.js in a fastpaced remote Agile environment
- Built and integrated RESTful APIs for data handling, user authentication, and business logic, using JWT for secure session management
- Resolved UI/UX bugs and improved cross-browser responsiveness using HTML5, CSS3, and JavaScript (ES6+)
- Managed database operations using MongoDB and SQL, including schema design and CRUD implementations
- Collaborated with senior developers via Git and GitHub — handling branching strategies, pull requests, and code reviews

PROJECTS

Fitness Tracker — Full-Stack Application — Personal Project

Jan 2026 – Feb 2026

- Planned and delivered a full-stack application across structured phases: requirements analysis, system design, development, testing, and project wrap
- Produced a work analysis report and process flow diagram documenting system architecture and milestone dates
- Built with Node.js, Express.js (REST APIs), MongoDB, and React.js with user authentication and workout logging
- Implemented a GraphQL API layer to enable flexible, client-driven data queries for workout and user data
- Managed all development independently, demonstrating self-motivation and ability to deliver to self-imposed deadlines
- Currently extending application capabilities with AI-powered features using LLM-based workflows.

Personal Portfolio — Frontend Project — Personal Project

2025

- Designed and developed a responsive single-page application using React.js and TypeScript, deployed on Vercel
- Implemented component-based architecture with React Hooks for state and lifecycle management
- Applied semantic HTML5, CSS3, and web accessibility (WCAG) best practices for cross-browser compatibility
- Focused on web performance optimization including lazy loading and fast load times

Karnel Travel — Full-Stack Booking Platform — Academic Project

2024

- Scoped and designed a multi-feature travel booking platform, mapping interdependencies between authentication, booking, and user management modules
- Produced database schema documentation and MVC architecture diagrams to communicate system design
- Implemented role-based authorization, relational database schema with SQL Server, and full CRUD operations
- Built using ASP.NET Core MVC — applying clean separation of concerns and maintainable architecture principles

Home Styler — Interior Home Design Application — International Hackathon Aug 2024

- Led a team of 4 in an international hackathon competing against 12 countries, winning 3rd place — demonstrating leadership, team coordination, and delivery under pressure
- Designed and built an interior home design SPA using React.js — scoping features, delegating tasks across the team, and managing a tight hackathon timeline
- Produced project documentation and UI architecture diagrams; implemented real-time state management, API integration, and cross-browser responsive design

AI-Driven Personal Fitness & Nutrition Portal v2.0 — Full-Stack Application — Personal Project 2026

- Developed RESTful APIs using FastAPI for user and fitness data management.
- Integrated TensorFlow.js and MediaPipe Pose for real-time posture detection.
- Leveraged Groq AI to generate personalized workout and nutrition plans.
- Implemented BMI, TDEE, calorie, and macronutrient calculations.
- Designed a SQLite database using SQLAlchemy ORM.
- Built a responsive React-based user interface with layered architecture.

EDUCATION

Associate's degree —Software Engineering — Aptech Learning

2024 – Present

- Database, OOP, API, Software Management, Software Development

CERTIFICATES & TRAINING

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- Corporate Strategy — University of London
 - Web Developer Internship Certificate — CodeAlpha
 - Programming in C# — Aptech Learning